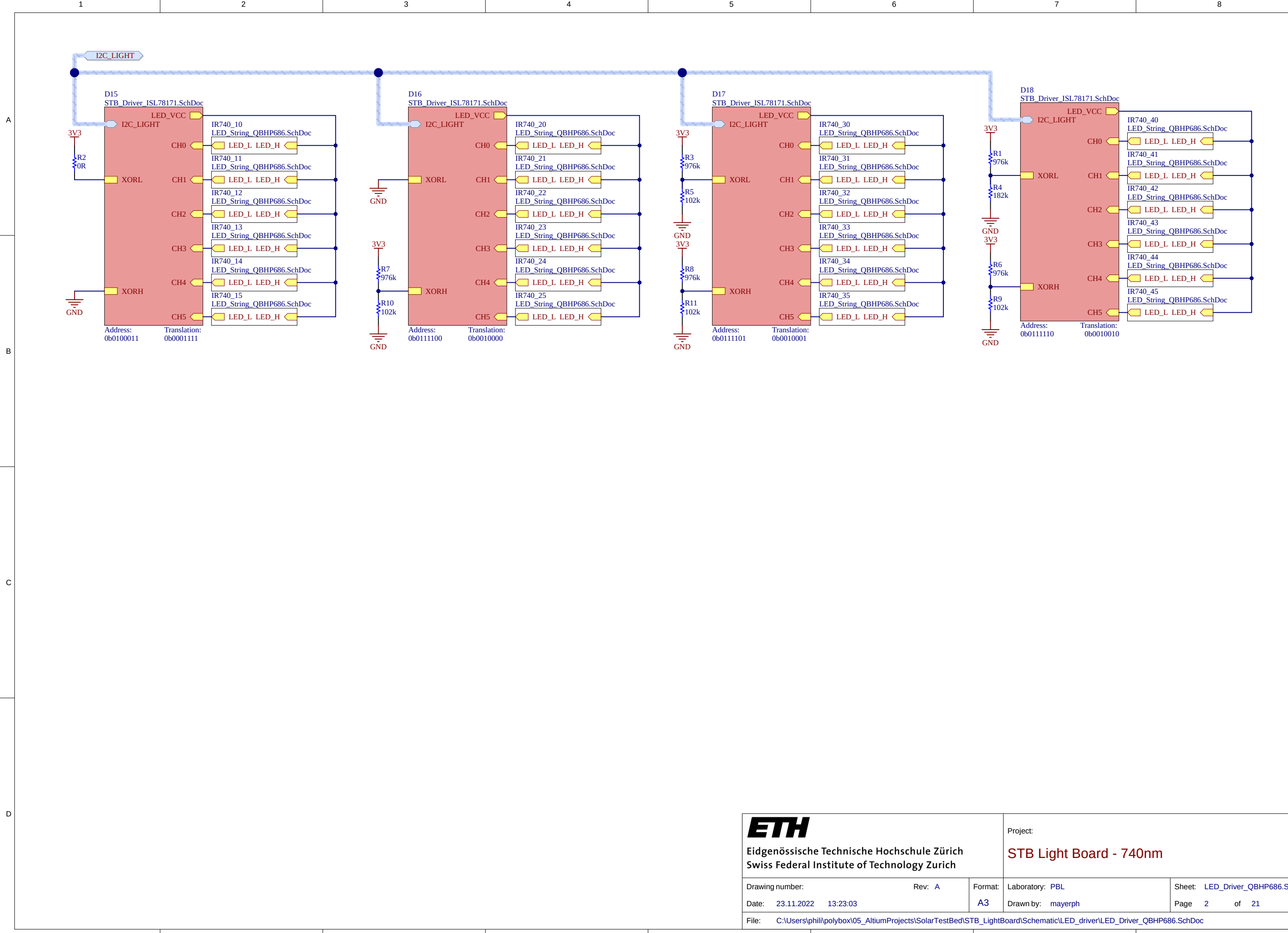
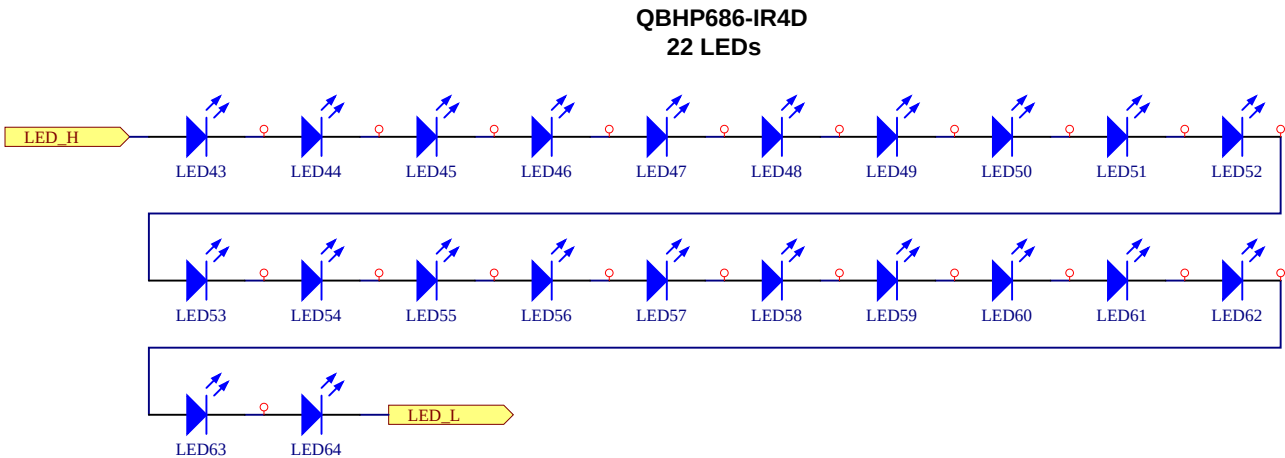






Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - 470nm String

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: LED_String_QBHP686.SchDoc

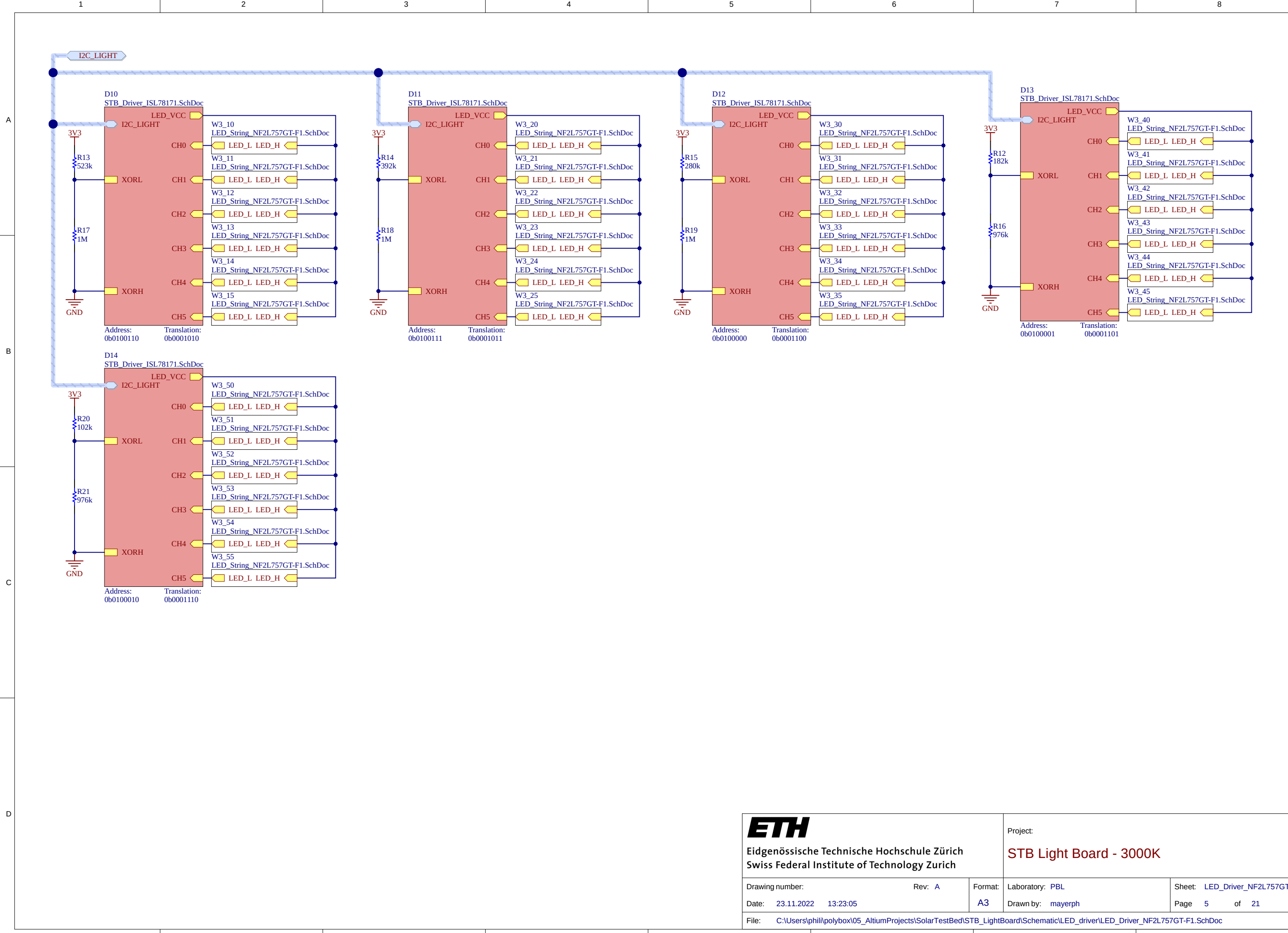
Date: 23.11.2022 13:23:04

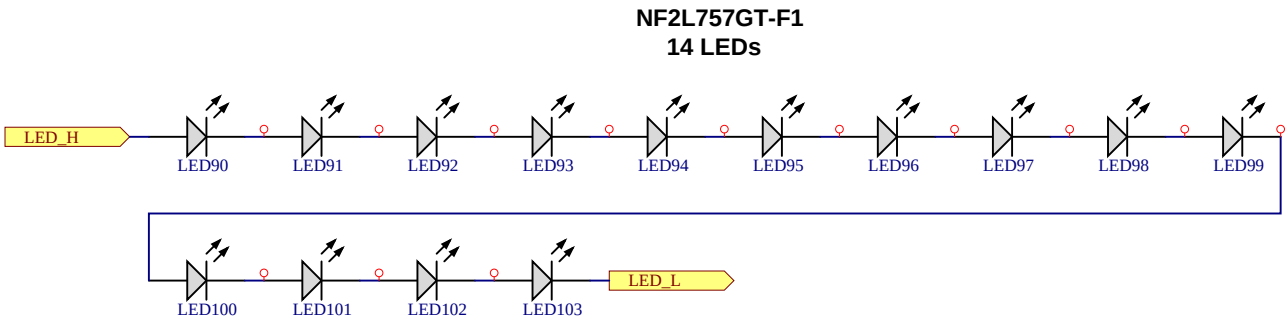
A4 Q

Drawn by: flmoll

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File: C:\Users\philip\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_string\LED_String_QBHP686.SchDoc





Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - 3000K String

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: LED_String_NF2L757GT-F1.S

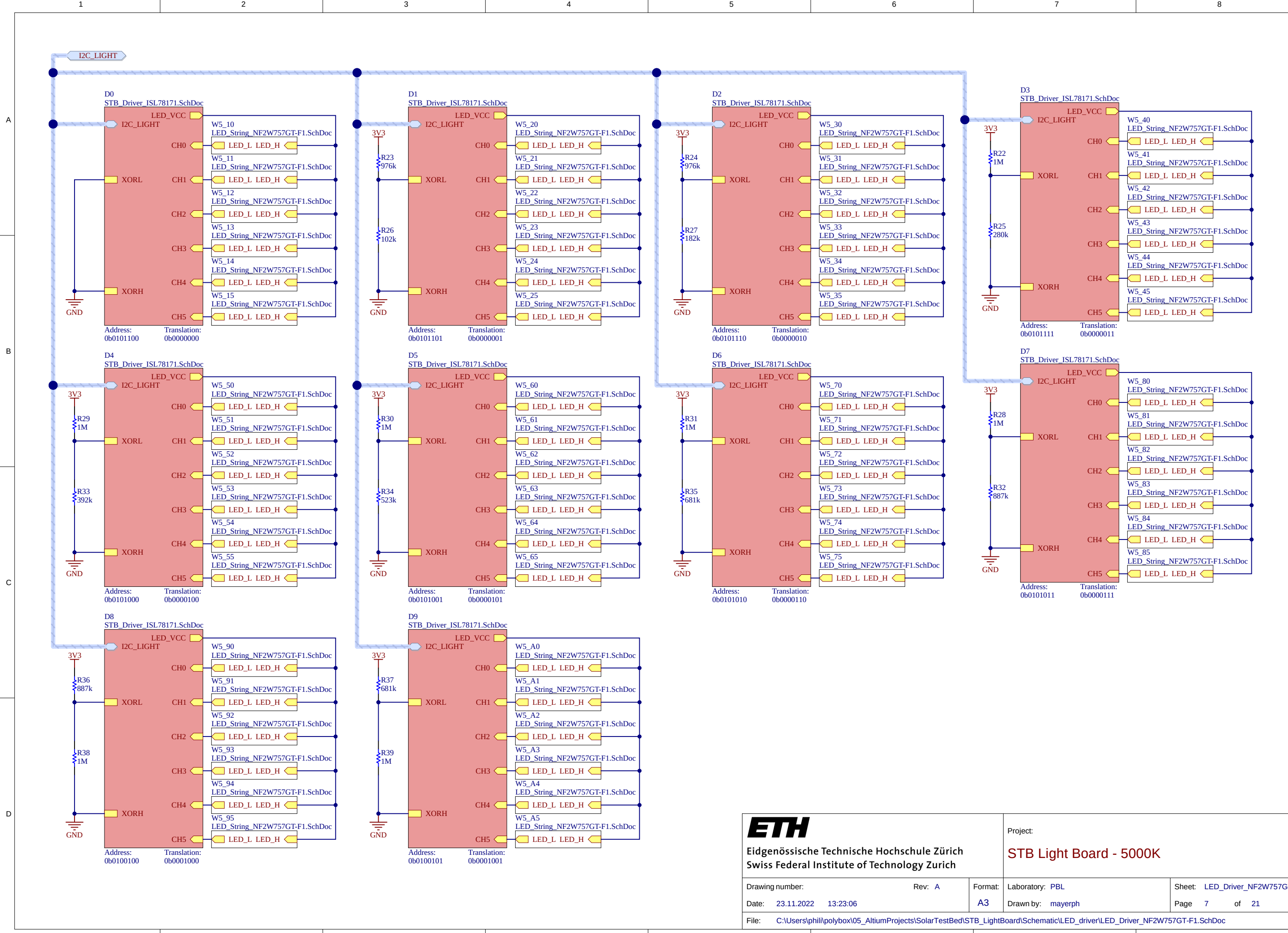
Date: 23.11.2022 13:23:06

A4 Q

Drawn by: flmoll

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File: C:\Users\phil\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_string\LED_String_NF2L757GT-F1.SchDoc

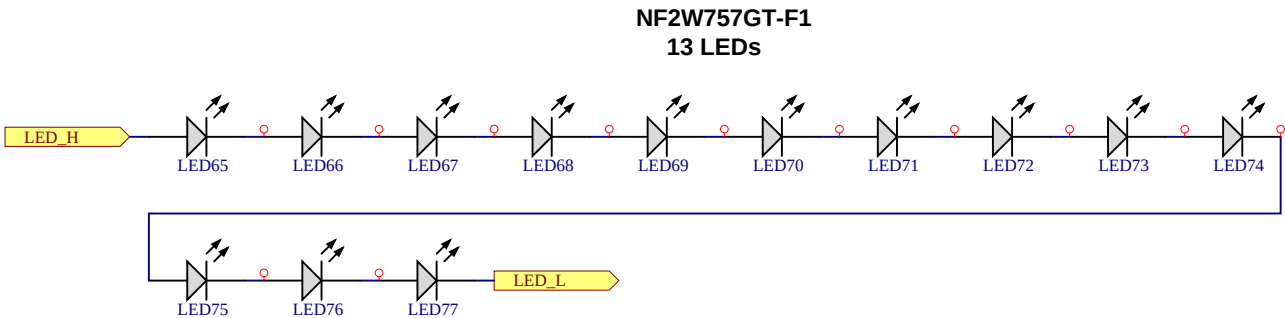




Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:
STB Light Board - 5000K

Drawing number:	Rev: A	Format:	Laboratory: PBL	Sheet: LED_Driver_NF2W757GT-F1.1
Date: 23.11.2022 13:23:06		A3	Drawn by: mayerph	Page 7 of 21
File: C:\Users\phill\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_driver\LED_Driver_NF2W757GT-F1.SchDoc				



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - 5000K String

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: LED_String_NF2W757GT-F1.S

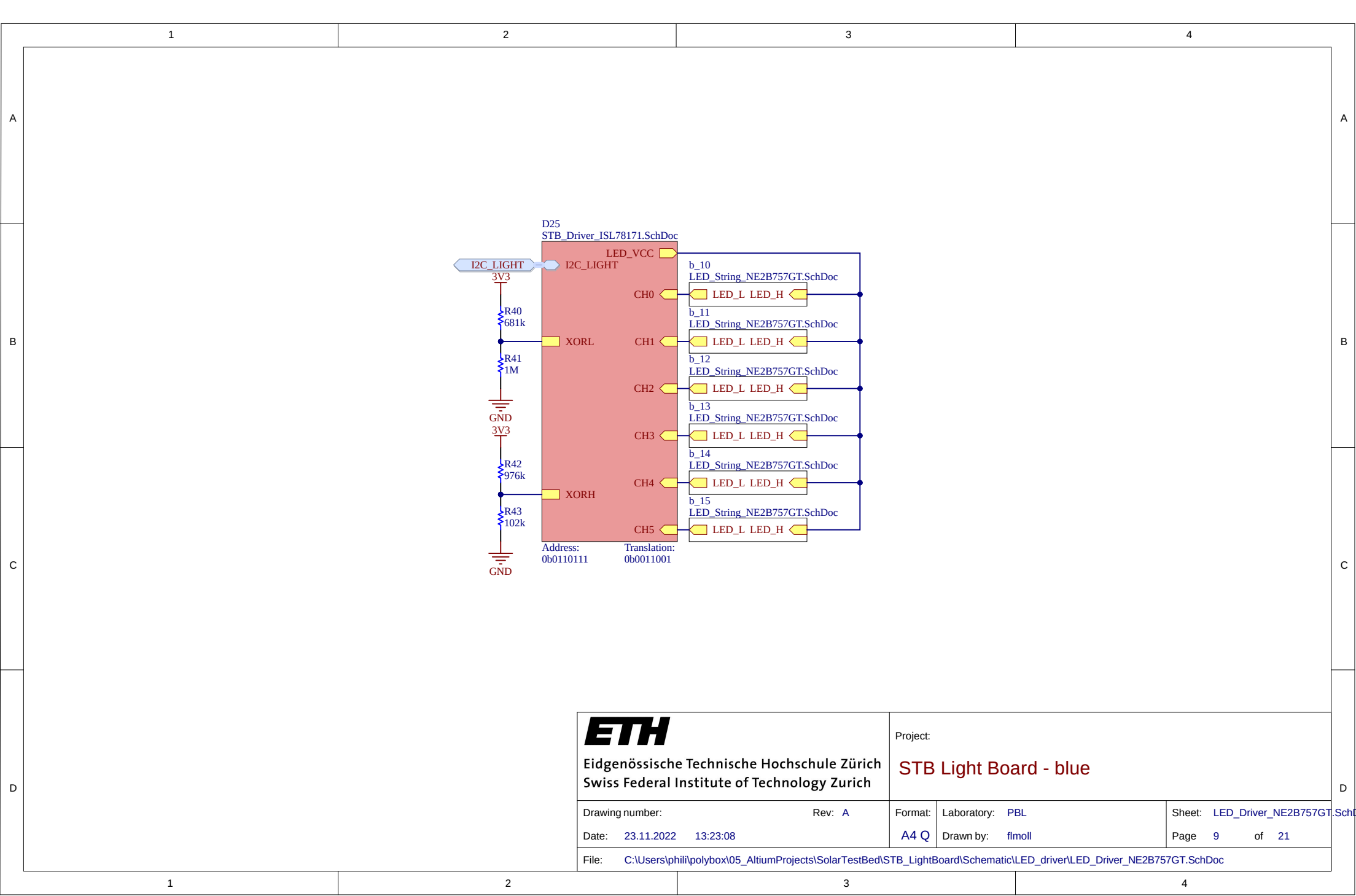
Date: 23.11.2022 13:23:08

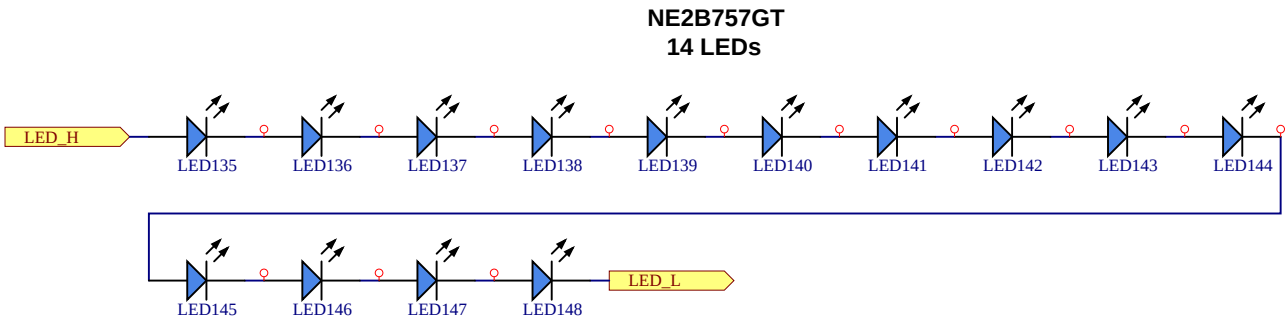
A4 Q

Drawn by: flmoll

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File: C:\Users\philip\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_string\LED_String_NF2W757GT-F1.SchDoc





Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - blue String

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: LED_String_NE2B757GT.SchDoc

Date: 23.11.2022 13:23:08

A4 Q

Drawn by: flmoll

Page 10 of 21

File: C:\Users\philip\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_string\LED_String_NE2B757GT.SchDoc

A

A

B

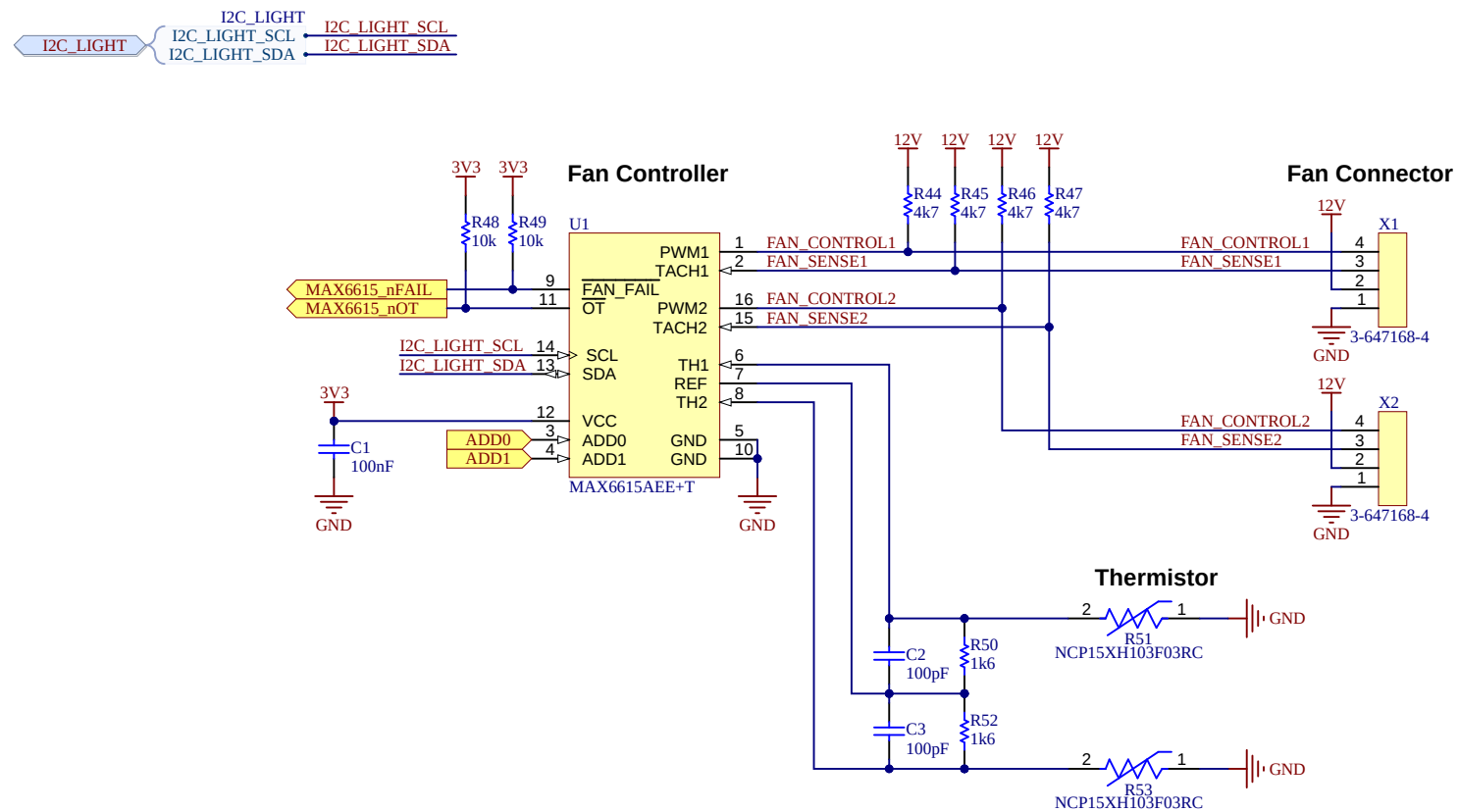
B

C

C

D

D

**ETH**

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - Fan Controller

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: STB_FanControl.SchDoc

Date: 23.11.2022 13:23:09

A4 Q

Drawn by: mayerph

Page 11 of 21

File: C:\Users\phill\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\STB_FanControl.SchDoc

1

2

3

4

A

B

C

D

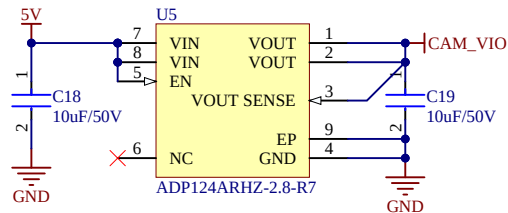
A

B

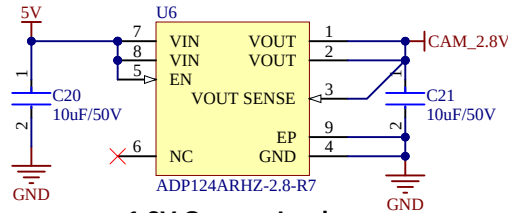
C

D

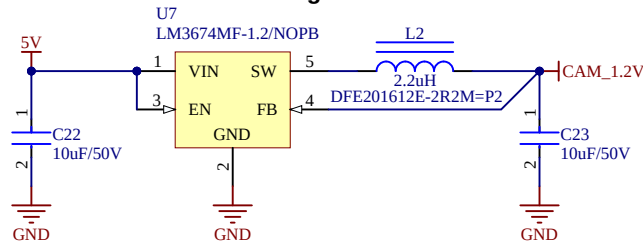
2.8V Camera IO



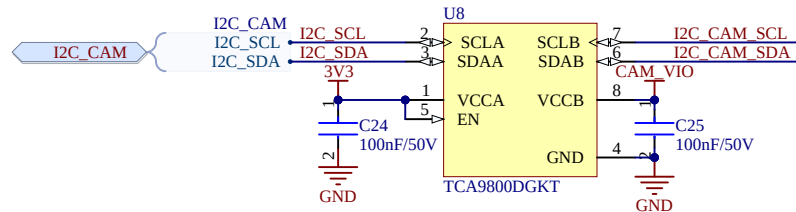
2.8V Camera Sensor



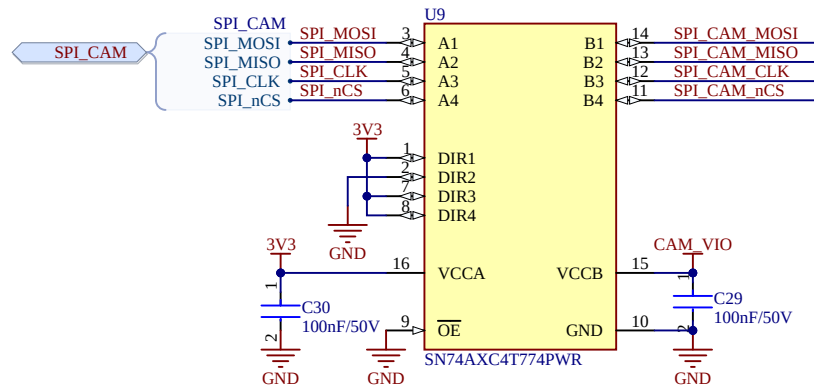
1.2V Camera Logic



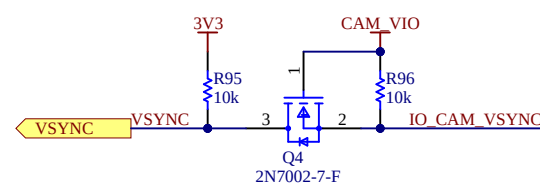
3.3V to 2.8V Translation



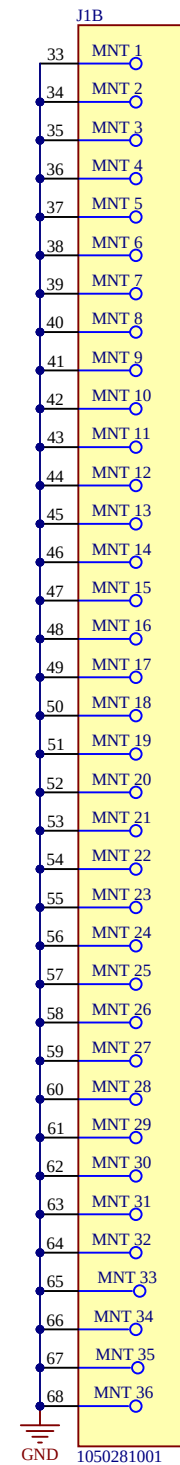
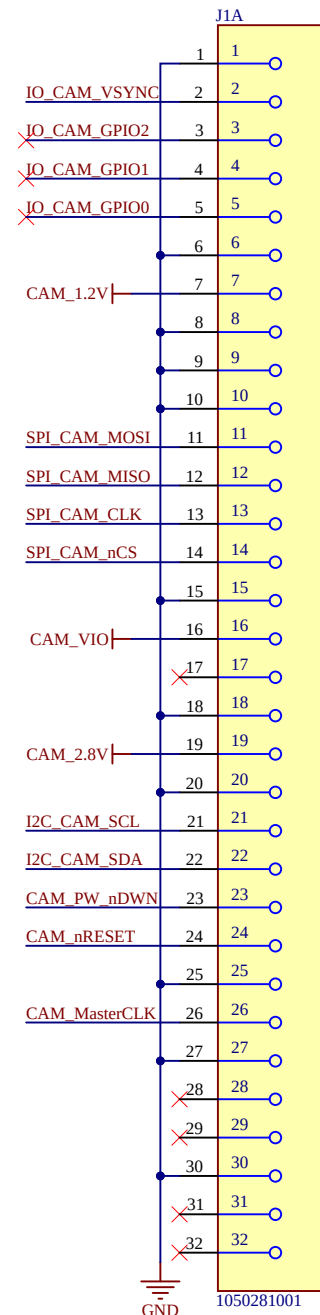
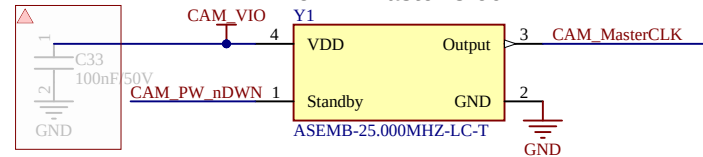
3.3V to 2.8V Translation



VSYNC Level Translation



25MHz Master Clock

**ETH**

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - Camera

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: STB_Camera.SchDoc

Date: 23.11.2022 13:23:09

A3

Drawn by: mayerph

Page 12 of 21

File: C:\Users\phillip\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\STB_Camera.SchDoc

1

2

3

4

5

6

7

8

A

A

B

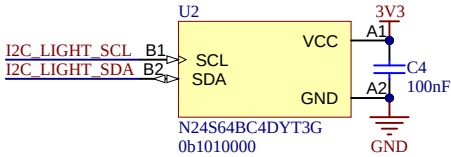
B

C

C

D

D



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - UID

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: STB_UID.SchDoc

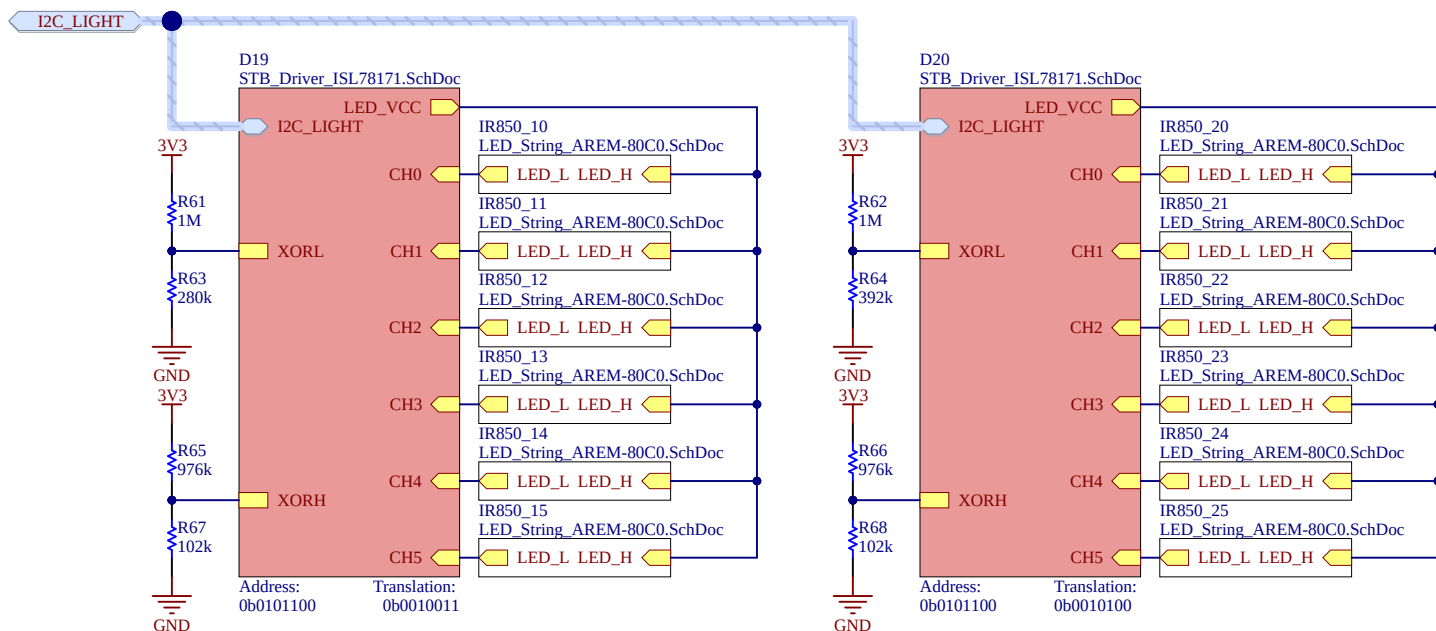
Date: 23.11.2022 13:23:10

A4 Q

Drawn by: flmoll

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File: C:\Users\phil\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\STB_UID.SchDoc



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - 850nm

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: LED_Driver_AREM-80C0.SchDoc

Date: 23.11.2022 13:23:10

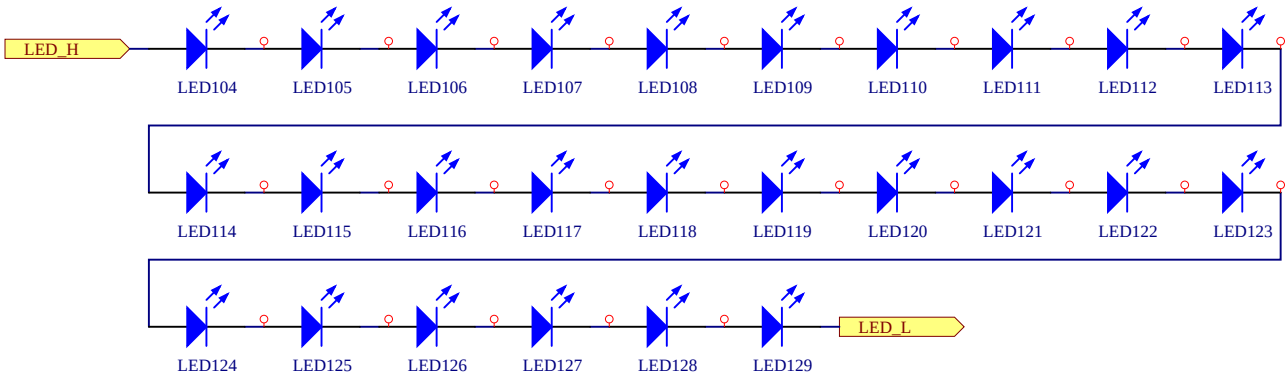
A4 Q

Drawn by: flmoll

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File: C:\Users\phill\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_driver\LED_Driver_AREM-80C0.SchDoc

AREM-80C0-LM000
26 LEDs



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - 850nm String

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: LED_String_AREM-80C0.SchDoc

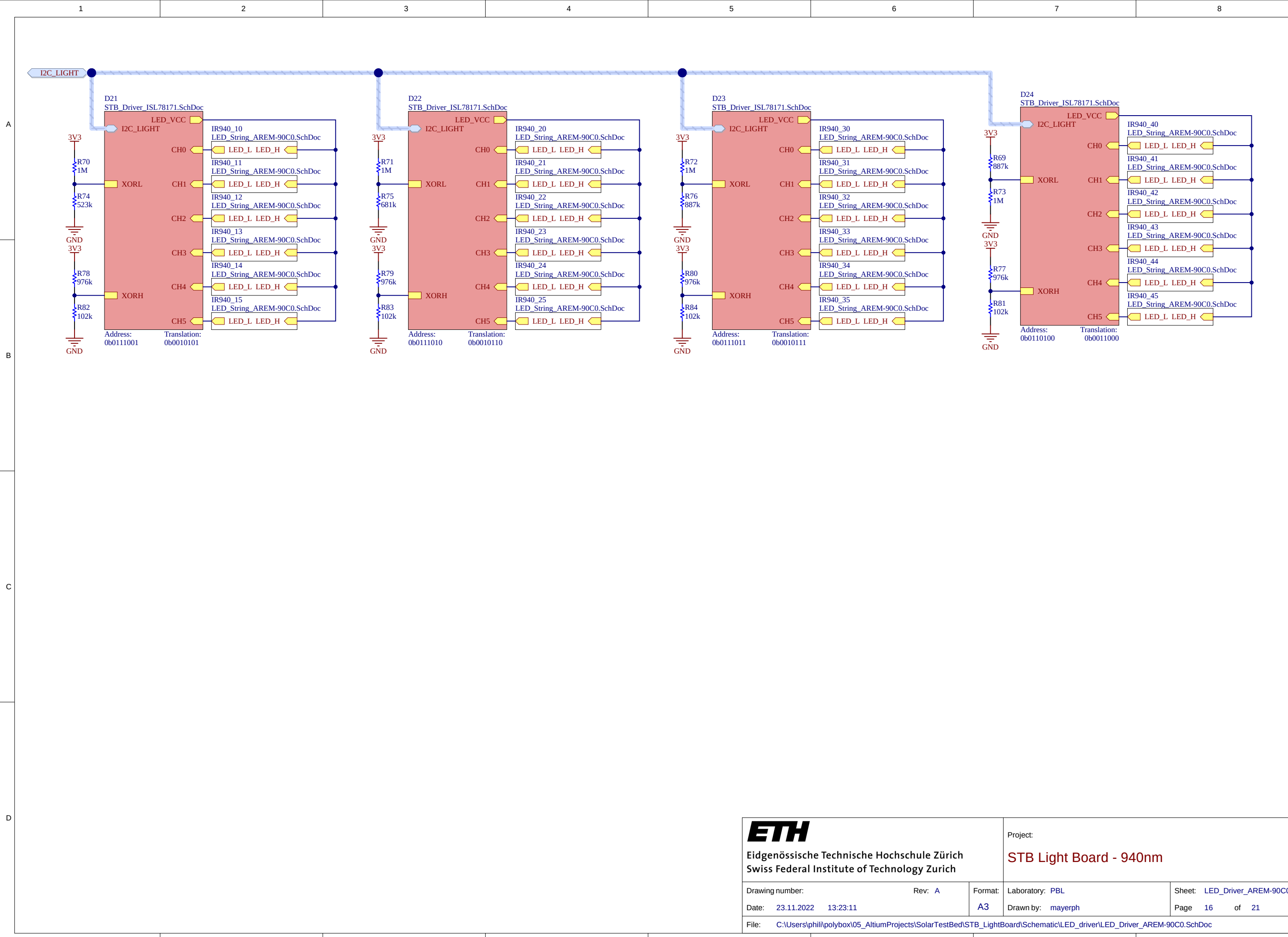
Date: 23.11.2022 13:23:11

A4 Q

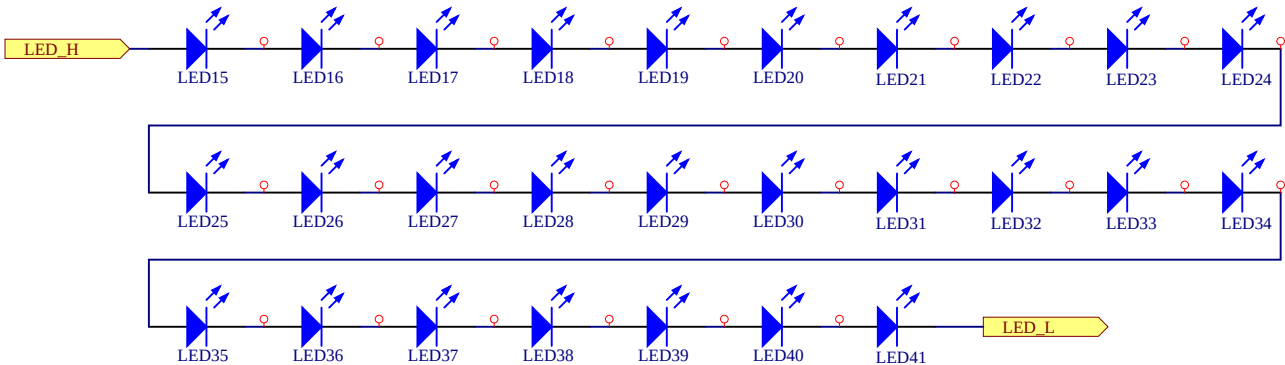
Drawn by: flmoll

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File: C:\Users\philip\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_string\LED_String_AREM-80C0.SchDoc



AREM-90C0-KL000
27 LEDs



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - 940nm String

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: LED_String_AREM-90C0.SchDoc

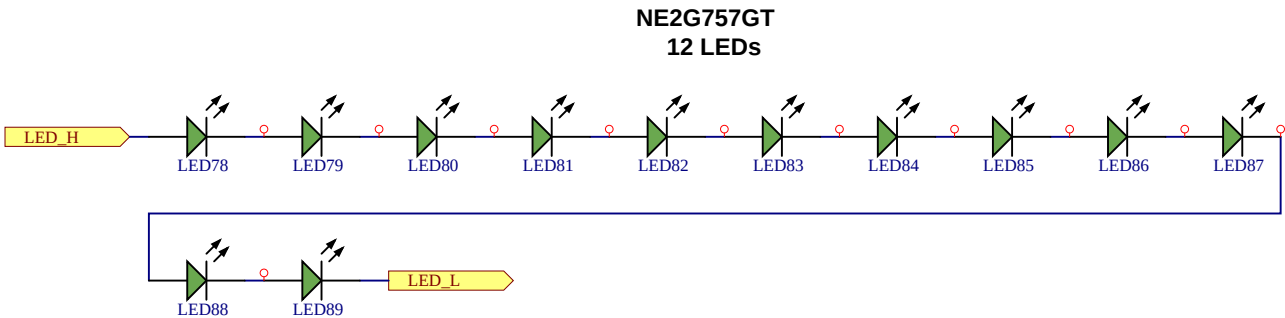
Date: 23.11.2022 13:23:12

A4 Q

Drawn by: flmoll

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File: C:\Users\phill\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_string\LED_String_AREM-90C0.SchDoc



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - green String

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: LED_String_NE2G757GT.Sch

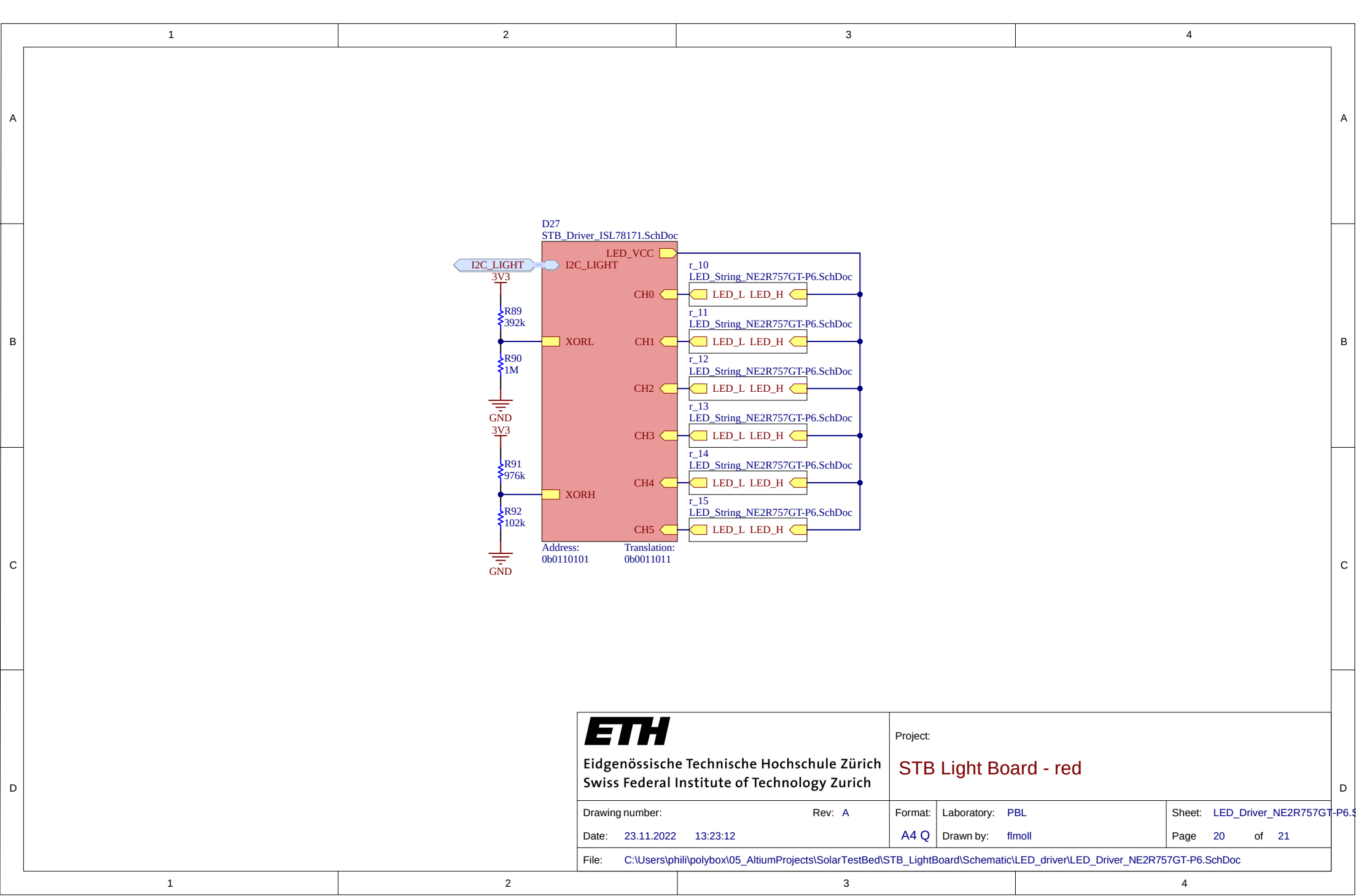
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A4 Q

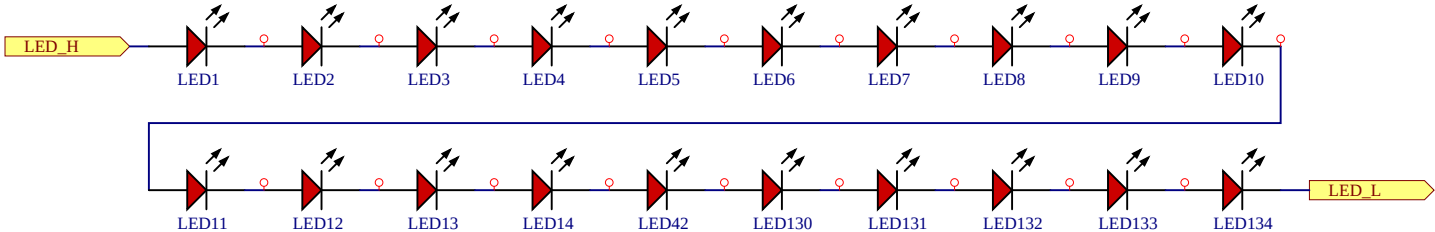
Drawn by: flmoll

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File: C:\Users\phill\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_string\LED_String_NE2G757GT.SchDoc



NE2R757GT-P6
20 LEDs



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

STB Light Board - blue String

Drawing number:

Rev: A

Format:

Laboratory: PBL

Sheet: LED_String_NE2R757GT-P6.S

Date: 23.11.2022 13:23:13

A4 Q

Drawn by: flmoll

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File: C:\Users\phil\polybox\05_AltiumProjects\SolarTestBed\STB_LightBoard\Schematic\LED_string\LED_String_NE2R757GT-P6.SchDoc